

ENTERPRISE AI INFRASTRUCTURE

CPU-to-GPU inference modernization

Fictional demonstration comparing CPU inference estate and GPU inference estate.

Fictional demonstration. Illustrative assumptions are not live vendor quotes, and modeled ROI is not a guarantee.

Executive summary

Current 5-year TCO	\$22,439,509
Proposed 5-year TCO	\$8,935,715
Modeled 5-year savings	\$13,503,794
Modeled productivity value	\$81,388
Modeled net value	\$13,585,182
Modeled ROI	152.0%
Simple payback	1.5 months
Assumption confidence	Low

Decision interpretation

Modeled financial advantage for the proposed infrastructure.

Modeled estimates are decision-support inputs, not guarantees or financial advice.

Scenario version	1
Analysis run	2e4052ff-288f-4d9e-94da-68797e42ac17
Generated	2026-07-21T06:31:14.160941+00:00
Calculation posture	Deterministic Decimal engine; no model performs financial calculations

Five-year comparison

Annual costs include visible recurring components and one-time transition costs.

Year	Current	Proposed	Delta
Year 1	\$2,649,055	\$1,320,327	\$1,328,728
Year 2	\$3,331,272	\$1,363,425	\$1,967,846
Year 3	\$4,218,153	\$1,659,953	\$2,558,200
Year 4	\$5,371,099	\$2,045,439	\$3,325,660
Year 5	\$6,869,929	\$2,546,571	\$4,323,359

Business-case measures

Three-year TCO savings	\$5,854,775
Five-year TCO savings	\$13,503,794
Five-year modeled net value	\$13,585,182
Five-year modeled ROI	152.0%
Break-even within five years	Yes

Scenario assumptions

All assumptions remain visible and should be replaced with approved evidence before a decision.

Workload envelope

Workload Type	inference
Model Size Billion	70.0
Training Runs Per Month	2
Monthly Requests Million	60.0
Average Demand Units	32.0
Peak Demand Units	85.0
Annual Growth Pct	30.0

Current and proposed states

Assumption	Current	Proposed
Label	CPU inference estate	GPU inference estate
Infrastructure Type	owned	owned
Accelerator Count	96	12
Compute Hourly Cost	2.4	4.8
Productive Utilization Pct	48.0	48.0
Storage Tb	180.0	180.0
Network Egress Tb Month	24.0	24.0
Power Kw	120.0	120.0
Pue	1.45	1.45
Staff Fte	2.5	2.5
Operating Hours Year	8760.0	8760.0

Transition and provenance

Migration cost	\$120,000
Implementation cost	\$65,000
Contract duration	3 year(s)
Sources	Fictional benchmark; Illustrative list pricing

Sensitivity analysis

One-factor deterministic ranges challenge utilization, price, growth, and energy assumptions. They are scenarios, not forecasts or probability distributions.

Dimension	Case	Assumption	Proposed TCO	Net value	ROI
Utilization	Low	38.0	\$10,608,039	\$11,912,857	112.3%
Utilization	Base	48.0	\$8,935,715	\$13,585,182	152.0%
Utilization	High	58.0	\$7,840,054	\$14,680,842	187.2%
Price	Low	4.32	\$8,479,422	\$14,041,475	165.6%
Price	Base	4.8	\$8,935,715	\$13,585,182	152.0%
Price	High	5.28	\$9,392,008	\$13,128,888	139.8%
Growth	Low	25.0	\$8,300,029	\$12,312,076	148.3%
Growth	Base	30.0	\$8,935,715	\$13,585,182	152.0%
Growth	High	35.0	\$9,628,587	\$14,972,815	155.5%
Energy	Low	0.104	\$8,577,335	\$13,943,562	162.6%
Energy	Base	0.13	\$8,935,715	\$13,585,182	152.0%
Energy	High	0.156	\$9,294,095	\$13,226,801	142.3%

Interpret sensitivity as decision risk. A favorable base case with a fragile downside still requires validation.

Calculation lineage

Selected material outputs are shown below. JSON and CSV exports retain the complete analysis snapshot and lineage collection.

Output	Formula	Derived value
current.tco_3_year	sum annual total for years 1-3	10198480.25
current.tco_5_year	sum annual total for years 1-5	22439508.58
proposed.tco_3_year	sum annual total for years 1-3	4343705.53
proposed.tco_5_year	sum annual total for years 1-5	8935714.9
comparison.savings_3_year	current 3-year TCO - proposed 3-year TCO	5854774.72
comparison.savings_5_year	current 5-year TCO - proposed 5-year TCO	13503793.68
comparison.savings_3_year_pct	3-year savings / current 3-year TCO x 100	57.41
comparison.savings_5_year_pct	5-year savings / current 5-year TCO x 100	60.18
comparison.productivity_value_3_year	sum productivity value per hour x downtime hours x 12 x growth	35910.0
comparison.productivity_value_5_year	sum productivity value per hour x downtime hours x 12 x growth	81387.9
comparison.net_value_3_year	3-year TCO savings + 3-year modeled productivity value	5890684.72
comparison.net_value_5_year	5-year TCO savings + 5-year modeled productivity value	13585181.58
comparison.roi_3_year_pct	3-year net value / proposed 3-year TCO x 100	135.61
comparison.roi_5_year_pct	5-year net value / proposed 5-year TCO x 100	152.03
comparison.payback_months	transition investment / cumulative modeled monthly operating benefit	1.5
comparison.break_even_within_5_years	payback months exists and is no more than 60	true
confidence.score	source coverage weighted 80% + contract coverage weighted 20%	14.8
confidence.source_coverage_pct	source coverage weighted 80% + contract coverage weighted 20%	3.51
confidence.contract_coverage_pct	source coverage weighted 80% + contract coverage weighted 20%	60.0
confidence.level	source coverage weighted 80% + contract coverage weighted 20%	Low
executive_summary.net_value_5_year	copy or interpret the corresponding modeled result	13585181.58
executive_summary.roi_5_year_pct	copy or interpret the corresponding modeled result	152.03
executive_summary.payback_months	copy or interpret the corresponding modeled result	1.5
executive_summary.recommendation	copy or interpret the corresponding modeled result	Modeled financial advantage for the proposed infrastructure.